

High-Speed Non-Contact Thermal Display Using Infrared Rays and Shutter Mechanism

Sosuke Ichihashi
sosuke@star.rcast.u-tokyo.ac.jp
Information Somatics Lab, RCAST,
The University of Tokyo
Bunkyo-ku, Tokyo, Japan

Arata Horie
arata.horie@star.rcast.u-tokyo.ac.jp
Information Somatics Lab, RCAST,
The University of Tokyo
Bunkyo-ku, Tokyo, Japan

Masaharu Hirose
hirose@star.rcast.u-tokyo.ac.jp
Information Somatics Lab, RCAST,
The University of Tokyo
Bunkyo-ku, Tokyo, Japan

Zendai Kashino
kashino@star.rcast.u-tokyo.ac.jp
Information Somatics Lab, RCAST,
The University of Tokyo
Bunkyo-ku, Tokyo, Japan

Shigeo Yoshida
shigeo@star.rcast.u-tokyo.ac.jp
Information Somatics Lab, RCAST,
The University of Tokyo
Bunkyo-ku, Tokyo, Japan

Masahiko Inami
drinami@star.rcast.u-tokyo.ac.jp
Information Somatics Lab, RCAST,
The University of Tokyo
Bunkyo-ku, Tokyo, Japan

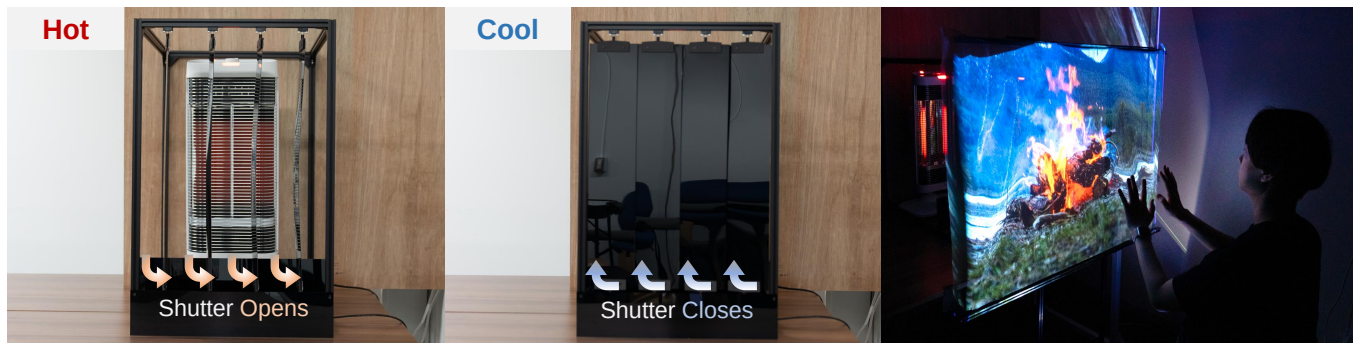


Figure 1: The user perceives a hot sensation as the shutter opens (left) and a cooling sensation as it closes (middle). The system could present thermal sensations without showing the shutter movements using an infrared transmissive mesh screen (right).

ABSTRACT

Considerable literature has explored the topic of haptics, including thermal presentation, in virtual reality (VR). However, the existing thermal presentation methods have limitations such as low responsiveness and need for contacts. As such, this study proposes a high-speed, non-contact thermal presentation method using infrared rays and a shutter mechanism consisting of fins. The results of the preliminary study suggested that the method could present temperature with different intensities according to the angle of the fin. Finally, several applications taking advantage of its features of high-speed and non-contact thermal presentation are proposed.

CCS CONCEPTS

• **Human-centered computing** → **Haptic devices.**

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
UbiComp-ISWC '21 Adjunct, September 21–26, 2021, Virtual, USA
© 2021 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 978-1-4503-8461-2/21/09...\$15.00
<https://doi.org/10.1145/3460418.3480160>

KEYWORDS

haptics; temperature; thermal feedback

ACM Reference Format:

Sosuke Ichihashi, Arata Horie, Masaharu Hirose, Zendai Kashino, Shigeo Yoshida, and Masahiko Inami. 2021. High-Speed Non-Contact Thermal Display Using Infrared Rays and Shutter Mechanism. In *Adjunct Proceedings of the 2021 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2021 ACM International Symposium on Wearable Computers (UbiComp-ISWC '21 Adjunct)*, September 21–26, 2021, Virtual, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3460418.3480160>

1 INTRODUCTION

There has been an increasing interest in haptics in VR. The senses of immersion and presence in VR increases as more sensations are presented in a realistic manner [1, 30]. The quality of visual and auditory stimulus presentation has improved considerably due to the emergence of head-mounted displays (HMDs) and spatial audio technologies and techniques. However, the presentation of other modalities such as haptic, gustatory, and olfactory sensations are still far from equivalent to what we perceive in physical space. Thus, they have potential to drastically increase the senses of immersion and presence.

Much research has been done in the field of haptics to increase the senses of immersion and presence. Haptic sensations are ones evoked by vibrotactile, pressure, skin stretch, kinesthetic, and thermal stimuli. A number of attempts have been made to reproduce various haptic sensations with devices stimulating these underlying sensations [4, 16, 23]. Although researchers have proposed several haptic devices using different haptic channels, the scale of research into thermal sensations is comparatively small. The reasons include the low responsiveness of existing presentation methods, which limits the range of temperature reproduction, and the fact that many of them are fixed to the environment or require contact because of their high power consumption, restricting the user's behavior or reducing the senses of immersion and presence.

Conventional presentation methods can be broadly classified into contact and non-contact methods. The former has relatively high responsiveness and is favored for high speed applications. However, it also requires contact between the device and user, which usually results in a reduced sense of immersion due to the constant awareness of the devices. The latter, on the other hand, does not require contacts to the devices, but the responsiveness and time resolution are low. A high-speed presentation without reducing the senses of immersion and presence is important for interactive applications including VR, but is difficult to achieve with the conventional methods. Thus, high-speed, non-contact thermal displays can drastically improve their experiences and can be integrated for real-time interactive applications.

Herein, we propose a method for high-speed, non-contact thermal presentation. We use a mechanical shutter consisting of four fins, which can rotate at 5.51 rad/s to control infrared radiant intensity from the infrared heater behind the shutter. The results of our preliminary study suggested the method could achieve rapid skin temperature changes.

In summary, this paper makes following contributions:

- (1) a novel method that can provide thermal sensation with high-speed and non-contact;
- (2) the results of our preliminary study suggesting the method could present temperature with different intensities according to the angle of the fins;
- (3) possible applications of the proposed method that take advantage of its features of high-speed and non-contact thermal presentation in Human-Computer Interaction (HCI) and VR.

2 RELATED WORK

Previous studies have explored various methods to present temperature. The existing methods can be divided into contact and non-contact methods.

In contact-type methods, thermal displays are directly placed on the body, and skin temperature is changed with conductions. Peltier elements, which transfer heat from one side to the other when voltage is applied, is one of the most prevalent contact-type displays due to its usability and availability [17, 24]. Although they are useful for localized thermal stimuli due to its compact size, they are not suitable for presentations on a large area. Additionally, Peltier elements have a limited responsiveness. Sato et al. achieved quicker responses by exploiting spatial summation and adaptation of thermal sensations with a set of four Peltier elements. Another popular

contact-type method is controlling fluid such as water [6, 25] and air flow [3]. This method generally has higher responsiveness and can present relatively rapid temperature changes. Other contact-type methods include resistive heating [28], visual-thermal interaction under an illusory body-ownership [13], gel packs [12], and thermal conductivity control [8]. Although contact-type thermal displays are widely used, the constant awareness and need of contacts possibly reduce the senses of immersion and presence.

In non-contact-type thermal presentation methods, on the other hand, skin temperature is usually changed with radiation or convection. Infrared lamps are widely used to remotely present thermal sensations [5, 10, 15, 26]. Han et al. combined infrared lamp, mist, water drop and air conditioning [7]. Other non-contact-type methods are air transportations [20, 22, 31], humidity adjustment [9], vaporization [21], visuo-thermal interaction [18, 19, 27], and olfactory [2]. These non-contact-type methods do not require contacts to the devices that can result in reductions in the senses of immersion and presence. However, their responses are slower than that of contact-type methods such as water flow.

These existing studies explored various methods and applications of thermal presentations. However, few have proposed methods and applications for high-speed, non-contact thermal presentation. This is due to a lack of method for high-speed control of the intensity of radiant or convective heat sources. In this study, we present a high-speed, non-contact thermal display using infrared rays and a mechanical shutter. Some researchers utilized a similar method to study the effect of thermal feedback during a fire evacuation training in VR [26]. However, it is unclear whether the method achieves high-speed thermal presentations and whether it can present in different intensities by controlling the shutter. To answer these questions, preliminary measurements are conducted (See details in Section 4). The possible applications of a high-speed, non-contact thermal display are also discussed.

3 DESIGN & IMPLEMENTATION

To present thermal sensation remotely, radiation and convection are usually used. To ensure quick responses, the travel time of heat energy between controlling units and human skin should be as short as possible. When radiation and convection are compared from this perspective, radiation is more suitable because infrared rays travel faster than the heat medium of convection. As the conventional approach of controlling infrared source does not change the radiant intensities rapidly, we decided to modulate the passing rate of infrared rays at the control unit between human and the infrared source. To control the passing rate, a shutter mechanism was used because it is easy to control and the rate can be continuously adjusted over a wide range. As such, we designed a high-speed, non-contact thermal display consisting of an infrared source and a mechanical shutter.

For an infrared source, we used a 1.15 kW infrared heater (CORONA DH-1220R), which emits infrared rays with wavelengths ranging from 3 to 20 μm without causing any air flow. We designed a mechanical shutter with four rotating fins. We chose polycarbonate as the material of the fins because of its high thermotolerance. Aluminium tapes with thickness of 0.09 mm were placed on one side of each fin to make sure infrared rays do not transmit the fins, and

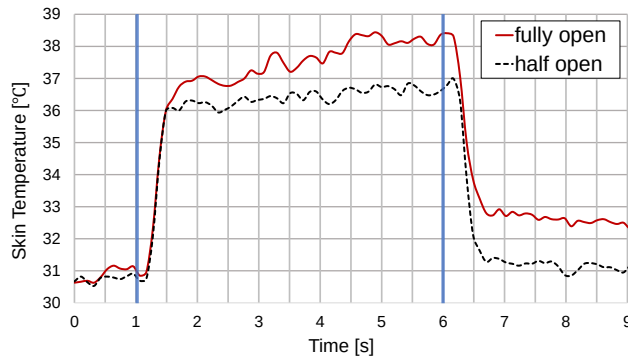


Figure 2: Skin temperature during fully and half open thermal presentations. The blue lines indicate the time at which control signals were sent.

they do not get too hot. Each fin is rotated using a servo motor (Futaba RS204MD), with torque of 0.21 N·m and angular velocity of 5.51 rad/s. Each fin is connected to the aluminium frame with a rotary bearing at one side and to the servo motor at the other side. The servo motors are controlled with a microcontroller (Arduino Uno R3). A polycarbonate plate with the aluminium tape on one side was placed to hide microcontroller and servo motors from the heater because they can easily get overheated otherwise. The width, height and depth of the device are 500 mm, 780 mm and 700 mm, respectively. The total wattage required is 1.15 kW.

To control thermal intensities, the system controls the fin angle and changes the gap width between fins. The system computes the gap width projected to the front surface w m from the fin angle θ rad with the following equation: $w = W \cdot (1 - \cos\theta)$, where W m is the width of the fin. The proposed system works solely with Arduino, so it can be easily controlled with different programs when used for applications.

4 PRELIMINARY MEASUREMENTS

To examine whether the system can realize high-speed thermal presentations with different intensities, skin temperature was measured during the presentations in two conditions: fully open and half open. In fully open condition, the shutter is initially fully closed. Then, it is fully opened to present thermal stimuli for 5.00 s until it is fully closed again. In half open condition, the shutter is half opened instead of fully opened. Here, half open means the gap width between fins w is halved from fully open condition. Thus, in half open condition, the fins rotate $\pi/3$ rad from the fully closed state, while they rotate $\pi/2$ rad in the fully open condition. For each condition, skin temperature was measured ten times on the cheek 0.80 m away from the heater with a thermography camera (TIM-03) with 9.50 Hz sampling rate. The room temperature was kept between 24 °C and 25 °C throughout the measurements. The interval between the trials was 60 s, and the trials were started after confirming the skin temperature gets back to the initial temperature measured before the experiment.

Figure 2 shows the measured temporal changes in the skin temperature. The plotted values are the mean of the ten measured values. Control signals to open and close the shutter were sent one

second and six seconds after starting measurements, respectively. Sharp rises and falls in skin temperature can be seen for both cases. In fully open condition, the mean skin temperature was 30.91 °C when the control signal was sent, and it rose to 35.99 °C in 0.50 s. it dropped from 38.41 °C to 33.25 °C in 0.50 s. In half open condition, the mean skin temperature rose from 30.69 °C to 35.99 °C in 0.50 s, and it dropped from 36.61 °C to 32.11 °C in 0.50 s. Additionally, there is a trend that the skin temperature is higher for fully open conditions than for half open conditions.

5 USER FEEDBACK

We gathered the feedback from 14 people who experienced the proposed system. The following are some of the comments we received.

Comments on Thermal Feedback

- I was surprised because the temperature change was much faster than I expected.
- I felt relatively cool when the shutter closes.
- It was warm and comfortable when the shutter is half opened, but it was hot and uncomfortable when the shutter is fully opened.
- The thermal presentation was so quick that I felt a kind of pressure.
- I felt uncomfortable with the rapid temperature changes.
- I felt as if the wind was blowing, since my only experience with rapid and non-contact temperature changes is with wind.

Comments on Overall Experience

- It was a new experience for me because I usually do not have a chance to feel frequent temperature changes like this.
- Hearing the servo motors move and seeing the fins rotate make the experience worse.

All the participants were able to perceive rapid temperature changes. Most of them also reported that they were able to perceive thermal sensations with different intensities depending on the fin angles. As some participants pointed out, the thermal presentation is accompanied by servo motor noise and visible fin rotations that may limit the applicability of the system. It is important to note that rapid temperature changes may lead to discomfort as reported by some participants. One unanticipated finding was that high-speed, non-contact thermal presentation might evoke other sensations such as pressure and wind.

6 DISCUSSION & APPLICATIONS

The objectives of the present study were to introduce a method for high-speed, non-contact thermal presentations and to give insights into its capabilities, especially its responsiveness and intensity adjustability. The results of the preliminary measurements and user feedback implied that the proposed method could have a high responsiveness and some intensity adjustability though presentations of cooling sensation require pre-heating. A note of caution is due here since the current study does not involve any psychophysical experiments to evaluate the system from the perspective of human perceptions. In addition, the perceived intensity will change according to the distance between the user and device. Therefore, further research should be undertaken to investigate the human perception aspects such as time to perceive temperature and perceived thermal

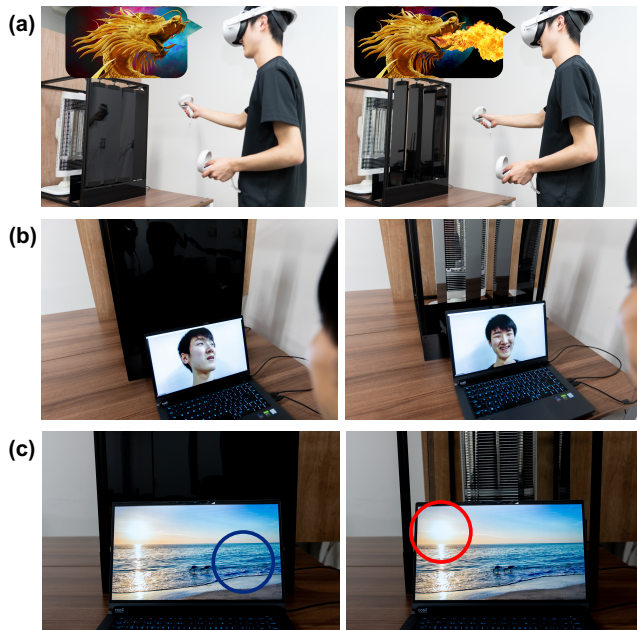


Figure 3: (a) Reproduction of Non-contact, Rapid Temperature Changes; (b) Gaze Presentation with Thermal Feedback during Remote Communications; (c) Gaze-based Thermal Feedback for Visual Media

intensities for different fin angles and distances between the user and device. Considering its applications in VR and HCI, further study on control models and systems is important. One interesting finding from the user study was that high-speed, non-contact thermal presentation might evoke other sensations such as pressure and wind. A possible explanation for the wind sensation might be that the sensation of non-contact rapid temperature changes is closely linked to wind sensation because they are usually caused by wind in our daily lives. The pressure sensation might be explained by the thermal discomfort caused by the rapid temperature changes. The discomfort could be interpreted as pressure. Another possible explanation for this is that the participant interpreted the aforementioned wind sensation as a pressure sensation. Further work is required to confirm and explain these observations. Last but not least, applications of high-speed, non-contact thermal display should be explored in VR and HCI research. Previous studies on thermal feedback mostly focused on static, predetermined, or slow thermal feedback due to the limitations of the thermal display. Therefore, there is abundant room for further progress in fast and interactive thermal feedback applications. Here we introduce and discuss possible applications of the proposed method that take advantage of its features of high-speed and non-contact thermal presentation.

Reproduction of Non-contact, Rapid Temperature Changes

One simple application is reproduction of non-contact, rapid temperature changes in VR or other media. As the system can present rapid temperature changes, which are hard to reproduce remotely with the existing methods, the senses of immersion and presence

could be improved in situations where temperature changes rapidly. Such scenarios include entering a warm room in cold weather, encountering fire breathing dragons, and being caught in explosions. The example is shown in Figure 3 (a).

Gaze Presentation with Thermal Feedback during Remote Communications

This system gives the user thermal feedback according to the gaze of the other user during remote communications. Thermal feedback is applied to the user when they are being watched by the other user through the screen as shown in Figure 3 (b). The more the other user's gaze is diverted from the user, the weaker the thermal presentation gets. It is reported that gaze presentations in remote communications improve communications and impressions of others [14]. As thermal feedback improves impressions of others and increases social proximities [11, 29], gaze presentations using thermal feedback could enrich remote communications and further improve impressions of others.

Gaze-based Thermal Feedback for Visual Media

The user perceives dynamically changing thermal feedback according to their visual attentions on the monitor. In doing so, the user can perceive heat from individual objects they are looking at in visual media. For example, the user perceives a hot sensation when they look at the sun, while they perceive a cooling sensation when they look at the other parts as shown in Figure 3 (c). As humans usually perceive light accompanied by heat from heat sources they see, and this system could virtually reproduce the unique glare with heat sources such as the sun and fire.

High-Speed, Non-Contact Thermal Presentation without Visual Cues

In response to the user feedback that visible fin rotations disturb the experience, a prototype of thermal display without visual cues was made. Since infrared rays are transmitted through a mesh screen, the system could present non-contact rapid temperature changes without any visual cues as shown in Figure 1 (right).

7 CONCLUSION

In this paper, we proposed a high-speed, non-contact thermal display using infrared rays and a mechanical shutter. The preliminary measurements and user feedback suggested that the method can present rapid temperature changes with some intensity adjustability. Further, we discussed possible applications of the system. The study was limited by the absence of psychophysical experiments to evaluate the system from the human perception perspective. In spite of its limitations, the study certainly adds to our understanding of thermal presentation methods. Further work needs to be done to establish a proper control model for the system to apply it to broad VR and HCI applications.

ACKNOWLEDGMENTS

This work was supported by JST ERATO (JPMJER1701). I would like to express my great appreciation to Dr. Sohei Wakisaka for his valuable suggestions during the development of this work.

REFERENCES

- [1] Doug A. Bowman and Ryan P. McMahan. 2007. Virtual Reality: How Much Immersion Is Enough? *Computer* 40, 7 (July 2007), 36–43. <https://doi.org/10.1109/MC.2007.257>
- [2] Jas Brooks, Steven Nagels, and Pedro Lopes. 2020. Trigeminal-based Temperature Illusions. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. ACM, Honolulu HI USA, 1–12. <https://doi.org/10.1145/3313831.3376806>
- [3] Shaoyu Cai, Pingchuan Ke, Takuji Narumi, and Kening Zhu. 2020. ThermoAir-Glove: A Pneumatic Glove for Thermal Perception and Material Identification in Virtual Reality. In *2020 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. IEEE, Atlanta, GA, USA, 248–257. <https://doi.org/10.1109/VR46266.2020.00044>
- [4] Alexandra Delazio, Ken Nakagaki, Roberta L. Klatzky, Scott E. Hudson, Jill Fain Lehman, and Alanson P. Sample. 2018. Force Jacket: Pneumatically-Actuated Jacket for Embodied Haptic Experiences. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. ACM, Montreal QC Canada, 1–12. <https://doi.org/10.1145/3173574.3173894>
- [5] H.Q. Dinh, N. Walker, L.F. Hodges, Chang Song, and A. Kobayashi. 1999. Evaluating the importance of multi-sensory input on memory and the sense of presence in virtual environments. In *Proceedings IEEE Virtual Reality (Cat. No. 99CB36316)*. IEEE Comput. Soc, Houston, TX, USA, 222–228. <https://doi.org/10.1109/VR.1999.756955>
- [6] Sebastian Günther, Florian Müller, Dominik Schön, Omar Elmoghazy, Max Mühlhäuser, and Martin Schmitz. 2020. Terminator: Understanding the Interdependency of Visual and On-Body Thermal Feedback in Virtual Reality. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. ACM, Honolulu HI USA, 1–14. <https://doi.org/10.1145/3313831.3376195>
- [7] Ping-Hsuan Han, Yang-Sheng Chen, Kong-Chang Lee, Hao-Cheng Wang, Chiao-En Hsieh, Jui-Chun Hsiao, Chien-Hsing Chou, and Yi-Ping Hung. 2018. Haptic around: multiple tactile sensations for immersive environment and interaction in virtual reality. In *Proceedings of the 24th ACM Symposium on Virtual Reality Software and Technology*. ACM, Tokyo Japan, 1–10. <https://doi.org/10.1145/3281505.3281507>
- [8] Seiya Hirai and Norihisa Miki. 2019. A Thermal Tactile Sensation Display with Controllable Thermal Conductivity. *Micromachines* 10, 6 (May 2019), 359. <https://doi.org/10.3390/mi10060359>
- [9] Kenta Hokoyama, Yoshihiro Kuroda, Ginga Kato, Kiyoshi Kiyokawa, and Haruo Takemura. 2017. Mugginess sensation: Exploring its principle and prototype design. In *2017 IEEE World Haptics Conference (WHC)*. IEEE, Munich, Germany, 563–568. <https://doi.org/10.1109/WHC.2017.7989963>
- [10] Felix Hülsmann, Julia Fröhlich, Nikita Mattar, and Ipke Wachsmuth. 2014. Wind and warmth in virtual reality: implementation and evaluation. In *Proceedings of the 2014 Virtual Reality International Conference*. ACM, Laval France, 1–8. <https://doi.org/10.1145/2617841.2620712>
- [11] Hans IJzerman and Gün R. Semin. 2009. The Thermometer of Social Relations: Mapping Social Proximity on Temperature. *Psychological Science* 20, 10 (Oct. 2009), 1214–1220. <https://doi.org/10.1111/j.1467-9280.2009.02434.x> Publisher: SAGE Publications Inc.
- [12] Dhruv Jain, Misha Sra, Jingru Guo, Rodrigo Marques, Raymond Wu, Justin Chiu, and Chris Schmandt. 2016. Immersive Terrestrial Scuba Diving Using Virtual Reality. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems*. ACM, San Jose California USA, 1563–1569. <https://doi.org/10.1145/2851581.2892503>
- [13] Shoko Kanaya, Yuka Matsushima, and Kazuhiko Yokosawa. 2012. Does Seeing Ice Really Feel Cold? Visual-Thermal Interaction under an Illusory Body-Ownership. *PLoS ONE* 7, 11 (Nov. 2012), e47293. <https://doi.org/10.1371/journal.pone.0047293>
- [14] Yuki Kinoshita, Masanori Yokoyama, Shigeo Yoshida, Takayoshi Mochizuki, Tomohiro Yamada, Takuji Narumi, Tomohiro Tanikawa, and Michitaka Hirose. 2017. Transgazer: Improving Impression by Switching Direct and Averted Gaze Using Optical Illusion. In *Proceedings of the 2017 ACM/IEEE International Conference on Human-Robot Interaction*. ACM, Vienna Austria, 53–62. <https://doi.org/10.1145/2909824.3020235>
- [15] A. Lecuyer, P. Mubuchon, C. Megard, J. Perret, C. Andriot, and J.-P. Colinot. 2003. HOMERE: a multimodal system for visually impaired people to explore virtual environments. In *IEEE Virtual Reality, 2003. Proceedings*. IEEE Comput. Soc, Los Angeles, CA, USA, 251–258. <https://doi.org/10.1109/VR.2003.1191147>
- [16] Robert W. Lindeman, Robert Page, Yasuyuki Yanagida, and John L. Sibert. 2004. Towards full-body haptic feedback: the design and deployment of a spatialized vibrotactile feedback system. In *Proceedings of the ACM symposium on Virtual reality software and technology - VRST '04*. ACM Press, Hong Kong, 146. <https://doi.org/10.1145/1077534.1077562>
- [17] Tomosuke Maeda and Tetsuo Kurahashi. 2019. TherModule: Wearable and Modular Thermal Feedback System based on a Wireless Platform. In *Proceedings of the 10th Augmented Human International Conference 2019*. ACM, Reims France, 1–8. <https://doi.org/10.1145/3311823.3311826>
- [18] George A. Michael, Hélène Galich, Solveig Relland, and Sabine Prud'hon. 2010. Hot colors: The nature and specificity of color-induced nasal thermal sensations. *Behavioural Brain Research* 207, 2 (March 2010), 418–428. <https://doi.org/10.1016/j.bbr.2009.10.027>
- [19] George A. Michael and Pauline Rolhion. 2008. Cool colors: Color-induced nasal thermal sensations. *Neuroscience Letters* 436, 2 (May 2008), 141–144. <https://doi.org/10.1016/j.neulet.2008.03.007>
- [20] Mitsuru Nakajima, Keisuke Hasegawa, Yasutoshi Makino, and Hiroyuki Shinoda. 2018. Remotely displaying cooling sensation via ultrasound-driven air flow. In *2018 IEEE Haptics Symposium (HAPTICS)*. IEEE, San Francisco, CA, 340–343. <https://doi.org/10.1109/HAPTICS.2018.8357198>
- [21] Mitsuru Nakajima, Keisuke Hasegawa, Yasutoshi Makino, and Hiroyuki Shinoda. 2021. Spatiotemporal Pinpoint Cooling Sensation Produced by Ultrasound-Driven Mist Vaporization on Skin. *IEEE Transactions on Haptics* (2021), 1–1. <https://doi.org/10.1109/TOH.2021.3086516>
- [22] Mitsuru Nakajima, Yasutoshi Makino, and Hiroyuki Shinoda. 2020. Remote Cooling Sensation Presentation Controlling Mist in Midair. In *2020 IEEE/SICE International Symposium on System Integration (SII)*. IEEE, Honolulu, HI, USA, 1238–1241. <https://doi.org/10.1109/SII46433.2020.9025959>
- [23] W.R. Provancher and N.D. Sylvester. 2009. Fingerpad Skin Stretch Increases the Perception of Virtual Friction. *IEEE Transactions on Haptics* 2, 4 (Oct. 2009), 212–223. <https://doi.org/10.1109/TOH.2009.34>
- [24] Nimesha Ranasinghe, Pravar Jain, Shienny Karwita, David Tolley, and Ellen Yi-Luen Do. 2017. Ambiotherm: Enhancing Sense of Presence in Virtual Reality by Simulating Real-World Environmental Conditions. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. ACM, Denver Colorado USA, 1731–1742. <https://doi.org/10.1145/3025453.3025723>
- [25] Masamichi Sakaguchi, Kazuki Imai, and Kyohei Hayakawa. 2014. Development of High-Speed Thermal Display Using Water Flow. In *Human Interface and the Management of Information. Information and Knowledge Design and Evaluation*, David Hutchison, Takeo Kanade, Josef Kittler, Jon M. Kleinberg, Alfred Kobsa, Friedemann Mattern, John C. Mitchell, Moni Naor, Oscar Nierstrasz, C. Pandu Rangan, Bernhard Steffen, Demetri Terzopoulos, Doug Tygar, Gerhard Weikum, and Sakae Yamamoto (Eds.), Vol. 8521. Springer International Publishing, Cham, 233–240. https://doi.org/10.1007/978-3-319-07731-4_24 Series Title: Lecture Notes in Computer Science.
- [26] Emily Shaw, Tessa Roper, Tommy Nilsson, Glyn Lawson, Sue V.G. Cobb, and Daniel Miller. 2019. The Heat is On: Exploring User Behaviour in a Multisensory Virtual Environment for Fire Evacuation. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. ACM, Glasgow Scotland UK, 1–13. <https://doi.org/10.1145/3290605.3300856>
- [27] Peter Weir, Christian Sandor, Matt Swoboda, Thanh Nguyen, Ulrich Eck, Gerhard Reitmayr, and Arindam Dey. 2012. BurnAR: Feel the heat. In *2012 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE, Atlanta, GA, USA, 331–332. <https://doi.org/10.1109/ISMAR.2012.6402599>
- [28] Reto Wettach, Christian Behrens, Adam Danielsson, and Thomas Ness. 2007. A thermal information display for mobile applications. In *Proceedings of the 9th international conference on Human computer interaction with mobile devices and services - MobileHCI '07*. ACM Press, Singapore, 182–185. <https://doi.org/10.1145/1377999.1378004>
- [29] L. E. Williams and J. A. Bargh. 2008. Experiencing Physical Warmth Promotes Interpersonal Warmth. *Science* 322, 5901 (Oct. 2008), 606–607. <https://doi.org/10.1126/science.1162548>
- [30] Bob G. Witmer and Michael J. Singer. 1998. Measuring Presence in Virtual Environments: A Presence Questionnaire. *Presence: Teleoperators and Virtual Environments* 7, 3 (June 1998), 225–240. <https://doi.org/10.1162/105474698565686>
- [31] Jiayi Xu, Yoshihiro Kuroda, Shunsuke Yoshimoto, and Osamu Oshiro. 2019. Non-contact Cold Thermal Display by Controlling Low-temperature Air Flow Generated with Vortex Tube. In *2019 IEEE World Haptics Conference (WHC)*. IEEE, Tokyo, Japan, 133–138. <https://doi.org/10.1109/WHC.2019.8816089>